

Part B: Design Justification

Windhoek HQ – Walvis Bay branch: Leased line / MPLS

This link is simulated as a serial WAN connection representing a leased line or MPLS circuit. The Windhoek–Walvis Bay corridor follows the B2 highway, one of Namibia's most developed routes, and is already served by real terrestrial fibre and MPLS infrastructure from Telecom Namibia and Paratus. Given both towns have commercial-density populations and existing operator backbone, a terrestrial link is the realistic, cost-effective choice — satellite would be unnecessary and more expensive for a route this well serviced.

Windhoek HQ – South Africa office: WAN link via submarine cable backbone

This link is simulated as a separate WAN connection, representing traffic routed over the regional submarine cable systems landing in Southern Africa (WACS, Equiano, SEACOM), rather than a single point-to-point terrestrial line. Namibia and South Africa are connected through overlapping cable infrastructure that gives multinational firms real redundancy options. The link is justified as a WAN/leased-line service riding this backbone capacity rather than a raw MPLS product, reflecting how Namibian operators actually reach South African points of presence.

Windhoek HQ – Global HQ (North America): VSAT / satellite link

This link is simulated using a Cloud-PT device representing a VSAT or LEO satellite connection. No realistic terrestrial or submarine path exists for direct simulation between Windhoek and a North American data centre at this scale, and satellite is the standard real-world method Namibian organisations use to reach distant or otherwise unreachable destinations. This mirrors Namibia's own domestic reliance on VSAT and newly licensed LEO services (such as the Twoobii-OneWeb rollout) for remote-area connectivity, applied here at global scale.

Windhoek HQ wireless LAN: Wi-Fi access point

The HQ site includes a wireless access point (SSID: KalahariHQ-WiFi, WPA2-secured) alongside its wired switch segment, reflecting a typical modern office where staff use both desktop PCs and mobile devices (laptop, smartphone). This satisfies the requirement for a functional wireless component at HQ while keeping the design realistic: a single flat /24 subnet serves both wired and wireless hosts, avoiding unnecessary VLAN complexity for a network of this size.

Addressing scheme rationale

Each site was allocated a distinct /24 subnet (10.10.0.0/24 through 10.40.0.0/24) to keep addressing simple to read and verify during marking, while WAN links use /30 point-to-point subnets (10.0.0.0/30, 10.0.0.4/30, 10.0.0.8/30) to avoid wasting address space on links that only ever need two usable hosts. Static routing was used throughout, since a four-site topology of this size does not yet justify the configuration overhead of a dynamic routing protocol such as

OSPF — though this would be the first change recommended if Kalahari Logistics' branch count were to double.